
Teaching Note

Negotiation Backtable Bots: Using GenAI to Improve Multiparty Negotiation Instruction

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The authors developed negotiation coaching bots using generative artificial intelligence (GenAI) and experimented with them in multiparty negotiation courses at MIT in the spring of 2024. This article focuses on one of the three kinds of bots we created: a coaching bot that allows learners to interact with the back tables of their assigned negotiation partners in an upcoming multiparty role-play simulation. Interacting with the back tables of other parties before entering into human-human negotiations can improve negotiation preparation in important ways. Interacting with backtable bots made very clear to students that negotiation parties are non-monolithic entities.

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We include descriptions of all the learning steps that our students went through, and quote them describing their reactions to interacting with a backtable bot and being coached by a GenAI negotiation coaching bot. We summarize our seven steps to building GenAI negotiation bots, which are applicable to all three types: preparation coaching bots, debriefing coaching bots, and backtable bots. We provide examples from our work on the backtable bots, and pay specific attention to the importance of prompt design. We conclude with recommendations for further development of negotiation coaching bots.

Keywords: multiparty negotiation, teaching, coaching, bots, artificial intelligence

Introduction¹

In 2023, negotiation educators and trainers began exploring ways of incorporating AI tools into their pedagogy (OpenAI 2023). The lead authors of this article—who have taught negotiation for more than five decades and have influenced the prevailing models of negotiation instruction in the fields of law, international relations, business management, public policy, and social work—were eager to capitalize on the potential of GenAI and set out to incorporate it into our teaching. In addition to teaching basic theories and techniques through the study of published scholarship on negotiation, we also emphasize close personal reflection by students on their role-play simulations, thereby increasing their awareness of the simulations' dynamics and improving their personal theory of negotiation practice.

Recent advances in the availability and capabilities of generative artificial intelligence (GenAI) have allowed us to develop and test negotiation coaching bots in multiparty negotiations, which are the focus of our engineering and urban planning negotiation courses. We created three kinds of GenAI bots: *negotiation preparation coaching bots* that students can use to prepare more effectively for upcoming in-person negotiations, *negotiation debriefing coaching bots* that students can use to reflect on and solidify the lessons they have learned from role-play simulations, and *negotiation backtable bots* that allow students to deepen their preparation by engaging virtually with colleagues or supervisors of the parties with whom they are negotiating. Based on the results of our experiments using these bots in two multiparty role-play simulations, we believe that backtable bots, in particular, are a game changer. Not

only do they enable students to manage unexpected conversations that might arise in real-life negotiations, but they enhance students' understanding of the multiple layers or rounds of their role-play simulations.

Essential Concepts in Pedagogy and Artificial Intelligence (AI)

We begin with descriptions of some concepts that are essential to an understanding of using GenAI to improve negotiation teaching.

Role-Play Simulations

A role-play simulation is different than a business school case study. While all the players in a role-play simulation receive common background information describing a specific context and moment in which the parties find themselves, the role players each receive separate confidential instructions. A case study, on the other hand, offers a shared narrative written from the standpoint of a historian or an observer. Students might be asked to imagine what they might do in the case study situation, but they do not receive confidential information directing them to pursue certain goals and not others. In role-play simulations, students are asked to take on the role of one of the parties in a difficult situation; they negotiate with other person(s) after reviewing their carefully crafted instructions. During the negotiation, each student knows what their party's goals are and how their performance will be evaluated by their teacher or instructor, sometimes using quantifiable scores. In a class of 24 students engaged in a multiparty negotiation involving six parties, we will assign students to four groups. They will play exactly the same game in four separate rooms. While the common and confidential instructions are the same for all role players, the negotiations in each room tend to evolve differently because of the way individuals play their assigned parts. Such in-the-moment, face-to-face experiences generate substantial real-time decision-making pressures as well as emotions that simulate what might happen in a real-life situation.

A simple illustration of a two-party negotiation is shown in [Figure 1](#). One student is assigned the role of a Buyer and another the role of a Seller. The two representatives meet at a negotiation table; their goal is to reach agreement on a specific question, issue, or deal.

A multiparty negotiation needs a much larger table, as shown in [Figure 2](#).

There are some key differences between two-party and multiparty negotiations that make preparing for and completing multiparty negotiations more difficult. For one, the emergence of alliances or coalitions is a major factor, affecting all aspects of the negotiation. Unlike in two-party negotiations, parties in multiparty negotiations must decide

Figure 1
The Negotiation Table in a 2-Party Role-Play Simulation

**At the Table
in a 2-Party
Negotiation**

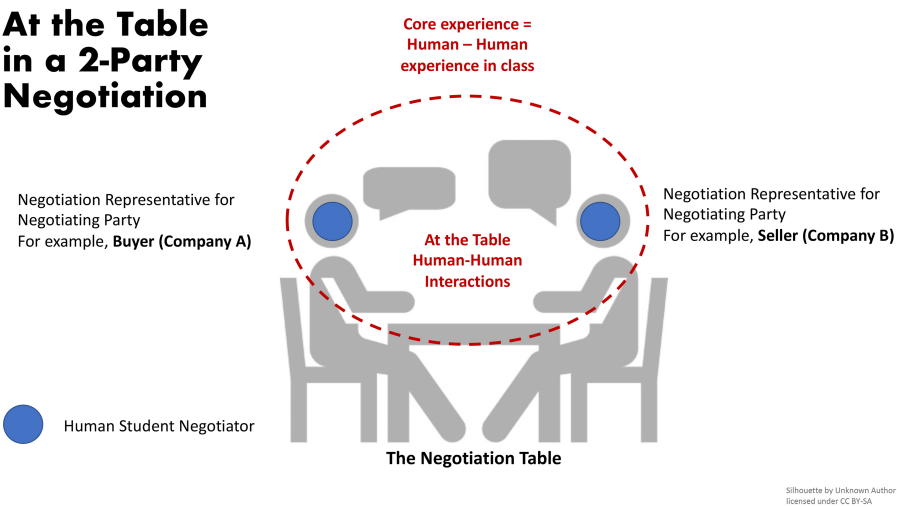
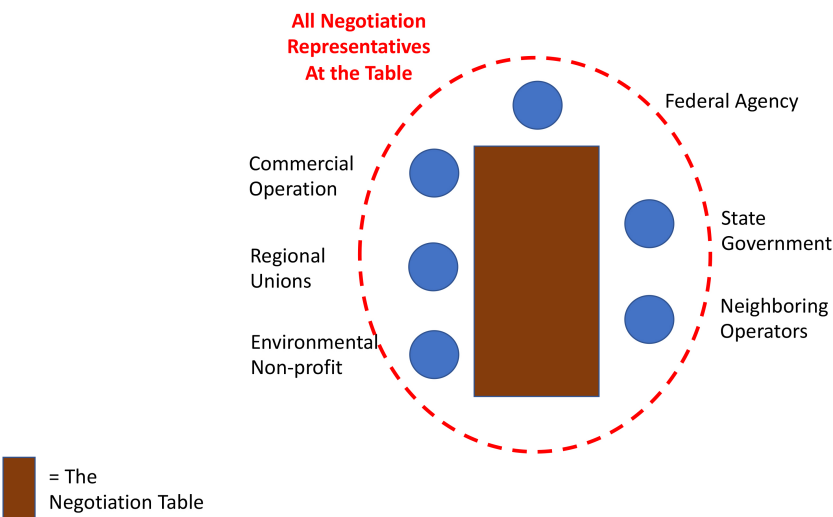


Figure 2
The Negotiation Table in a Multiparty Role-Play Simulation

Multiparty Negotiation At the Table

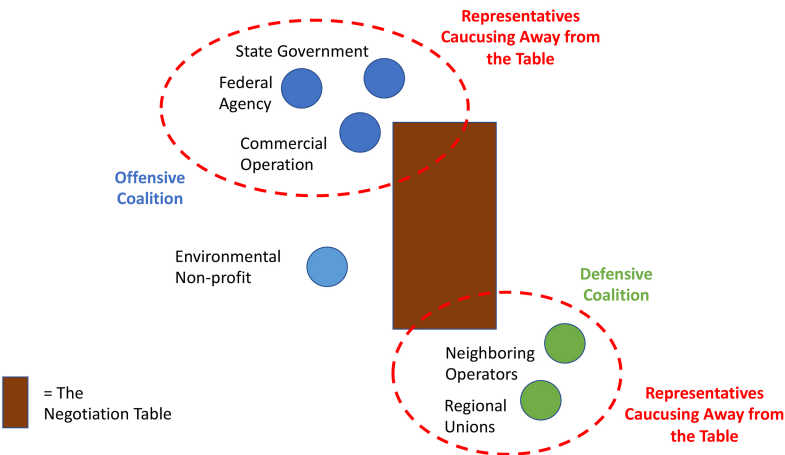


whether to join or form a coalition and whether to use a coalition to build support for a particular agreement or to block a worst-case agreement. They also must decide whom to talk to and in what order, and when and how to state their commitments (Olekals et al. 2003; Kozina et al. 2020). What are the ground rules and who enforces them? Is there a neutral facilitator or does one of the parties have permission to manage the conversation? Moreover, in multiparty negotiations, parties can pass private messages between them while at the table through notes, text messages, etc.

In a two-party negotiation, both parties are at the table all the time. In multiparty negotiations, the parties may meet all together (see Figure 2) and reach complete agreement on one or several issues. But if the decision rule specified in the simulation requires a majority vote to conclude the negotiation, the parties may break up into small groups, as shown in Figure 3. Three negotiators may caucus at one side of the room and try to form a winning or an offensive coalition by getting one more vote to join them for a majority. Meanwhile, two negotiators may caucus at the other side of the room and seek to build a defensive coalition by getting one more vote to oppose the emerging majority. A multiparty negotiation may involve a sequence of all-at-the-table segments and caucusing segments, and these may be carefully orchestrated

Figure 3
Using Caucuses and Building Coalitions in a Multiparty Negotiation

Multiparty Negotiation - Coalitions and Caucusing



or happen spontaneously. Finally, due to the ever-changing coalitional formations during a multiparty negotiation, there isn't a clear "no agreement" outcome spelled out in the parties' confidential instructions. This makes it hard to summarize "the value" of no agreement.

Negotiation Coaching

A negotiation coach works with a party to help them understand the specifics of their negotiation and to encourage them to prepare thoroughly for their upcoming interactions (Bachkirova and Clutterbuck 2010). Negotiation preparation usually includes clarifying a party's negotiation objectives, understanding their BATNA (best alternative to a negotiated agreement) (see, e.g., Sebenius 2017), identifying their personal interests, and trying to anticipate the interests of the other sides (Liu and Chai 2011). While negotiation coaches cannot guarantee good outcomes, they can materially improve preparedness by asking sequences of clarifying questions. Working with a negotiation coach enables learners to approach each new scenario with confidence and a strategic mindset. Research shows that good preparation can contribute to over 80% of the value realized in a negotiation.

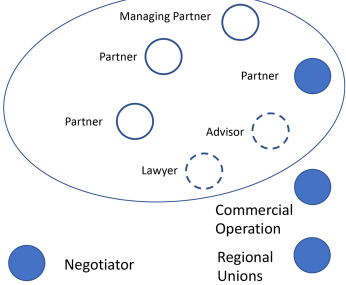
Back Tables in Negotiation

The designated negotiators sitting at the table are the "frontline" players (as in Figures 1 through 3). But in most cases, each negotiator represents an organization or a group that includes many people who do not appear at the negotiation table (for example, the negotiator's boss or key associates). These people are also stakeholders (and sometimes even decision-makers). We call them collectively the "back table." Before and after sitting down at the negotiating table, each negotiator may have had several conversations with members of their own back table; these are sometimes called "internal negotiations." Two examples of a back table are illustrated in Figure 4. One side of Figure 4 shows the Environmental Nonprofit, which is a partnership; the other side shows the Commercial Operation Company, which is a corporation. The designated negotiator for the Environmental Nonprofit is one of the organization's partners. The designated negotiator for the Commercial Operation Company is the company's business developer, who reports to the vice president of business development, who reports to the COO, who reports to the CEO. As per corporate policy, the business developer is authorized to negotiate, but any deal needs approval from others in the company. Back table stakeholders play a critical role in many negotiations by stipulating targets, approving informational inputs, specifying priorities, and setting deadlines. Internal negotiations with one's back table advisors and bosses often define the key parameters for an upcoming negotiation. When back table stakeholders from one side of a negotiation talk to back table stakeholders from another party, a "back

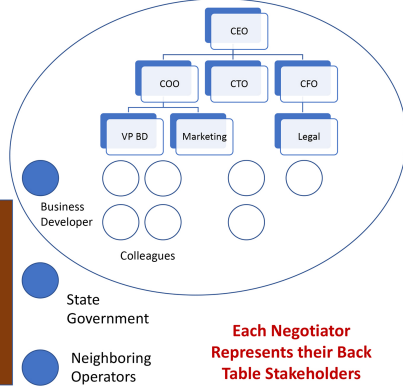
Figure 4
Back Table Stakeholders for 2 of the 6 Parties

Back Tables in Negotiation

Environmental Non-Profit's Back Table



Commercial Operation's Back Table



Each Negotiator
Represents their Back
Table Stakeholders

(illustrated for 2 of 6 parties)

channel” is created between knowledgeable persons who are not formally negotiating (since they are not the designated representatives), and these back-channel conversations have often enabled breakthroughs in, and added value to, the at-the-table formal negotiations (Wanis-St. John 2006). By engaging with the other sides’ back table stakeholders, these behind-the-scenes negotiators can sometimes explore “out-of-the-box” trade-offs and creative solutions in a protected setting (PON 2025). Most confidential instructions for negotiators in role-play simulations summarize what their back table wants from them, but provide no way for an at-the-table negotiator to talk to the other parties’ back tables.

Technology for Teaching Negotiation

Negotiation faculty have been at the forefront of using many types of technologies to enhance their pedagogy, including tools such as avatars and online platforms, which have improved learning and teaching results (Dinnar et al. 2021). Online learning platforms allow instructors to streamline and automate the process of managing role-play simulations (i.e., assigning roles, communicating instructions, managing the timing of group interactions, recording scores, and displaying results) both for in-person negotiations and online negotiations (video, voice, or text-based).²

Technologies for Developing Bots

Artificial intelligence (AI) refers to the ability of machines or computers to perform tasks that typically require human intelligence, such as problem-solving, decision-making, and understanding language (Mason 2003). It is also a subfield of computer science (Bemejo and Juiz 2023).

Generative artificial intelligence (GenAI) is a subset of artificial intelligence that employs advanced machine learning models to create novel and realistic content across various modalities, including text, images, audio, and video (Banh and Strobel 2023). These systems learn patterns and structures from vast datasets, enabling them to generate original outputs that mimic and extend beyond their training data.

Large language models (LLMs) are a type of artificial intelligence system designed for natural language processing tasks. They are typically deep neural networks with a large number of tunable parameters, trained on vast amounts of text data to learn patterns in language (Mitchell 2024). GPTs are a specific type of LLM.

Generative pre-trained transformers (GPTs) are built on the foundation of transformer architecture to understand words by looking at the previous words in a sentence or paragraph (Vaswani et al. 2017). The result is a type of artificial intelligence that can generate human-like text and content. In our work we will also refer to them as models or GPTs, adding specificity where we feel it is necessary.

Negotiation Coaching Bots

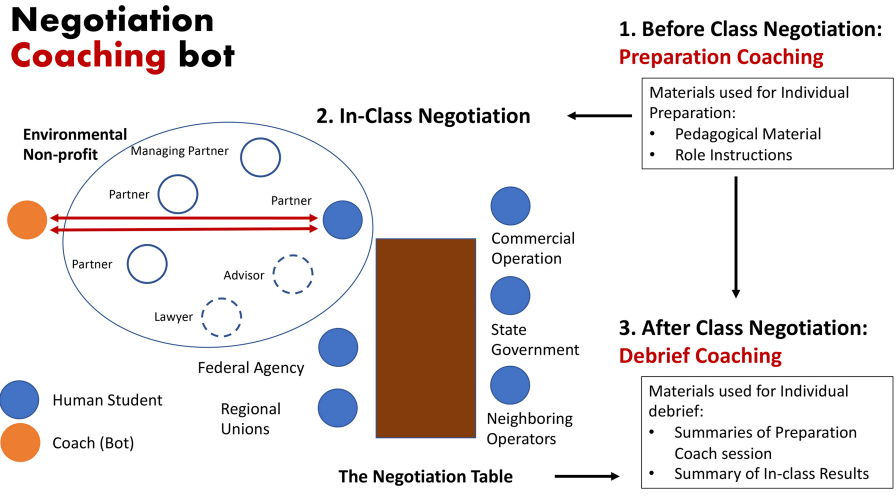
A coaching bot is an application, interface, or program that simulates interactions between a negotiation coach and multiple players (one at a time) in a role-play simulation. It does so in a way that is typical of the usual interactions between a college instructor and a student in a graduate negotiation class as that student gets ready for an assigned role-play simulation. It falls under the larger umbrella of chatbots, but with a clear distinction—it is only used for negotiation coaching, and functions only in a Socratic fashion. The coaching bot helps the learner see the choices they face and the possible consequences of responding poorly to the other sides' demands, threats, and false enticements (Shea et al. 2024). The negotiation coaching bots we have designed ask multiple rounds of directed questions, never telling the negotiator what to do. We have built several types of coaching bots:

- *Preparation coaching bot:* a negotiation coach that focuses exclusively on preparing a human negotiator in advance of a specific assigned role-play simulation. In Figure 5, step 1 shows a negotiation coaching bot interacting with a student in a 6-party role-play negotiation simulation in preparation for the student's in-person (human-to-human) in-class negotiation (step 2).
- *Debrief coaching bot:* After the in-class assigned negotiation (step 2 in Figure 5), we have developed a debriefing coaching bot to debrief each student's human experience during the negotiation. The debriefing coaching bot has access to the student's strategy going into the negotiation (i.e., the results of their interaction with the

preparation bot) as well as to the actual results of the negotiation, which have been fed into the learning platform by the students or a teaching assistant. Although the student may experience a feeling of continuity in talking to the “same” coaching bot that gave them help during their negotiation preparations, in reality, the debriefing is implemented using a different bot. After the in-class debrief at the conclusion of the role-play simulation, the student may continue to interact with the debriefing bot (i.e., one on one) to find out more about what the other roles were thinking and what they might have done differently. The debriefing bot now has access to all roles and game design options.

- Backtable negotiation bot:** a different kind of bot we developed that does not coach using the Socratic method, but rather is designed to help each student by “role-playing” a person from one of the other parties’ back tables before their scheduled role-play simulation. This is similar to what a human negotiation consultant (i.e., a human coach) might do when preparing an executive for a real-world negotiation, asking questions such as: “Let’s assume that I am someone from the opposing party’s back table, what questions would you like to ask me? Let’s role-play that, how would you ask your questions?” A backtable bot holds a very specific chat conversation with each student to help deepen the student’s negotiation preparation.

Figure 5
One Student’s Negotiation Coaching Bot Interactions in a 6-Party Negotiation



Personal Theory of Practice (PTOP)

The term “personal theory of practice” (PTOP) in negotiation refers to an individual’s unique framework for anticipating, understanding, and responding to an emerging negotiating situation (Dinnar and Susskind 2019). It usually represents an individual’s effort to integrate readings, past personal reflections, and beliefs into the principles they choose to follow in practice. PTOP emphasizes the importance of self-awareness and the need to continuously develop one’s own negotiation skills through reflections on practice. Changing a student’s PTOP in a negotiation class involves reshaping the internal mindset or frameworks they use to determine when and how to act or respond to specific cues or challenges. Asking students to articulate their PTOP requires building trust, exposure to and discussion of theoretical models, opportunities to try them out and reflect on what happened compared to what they expected, and a self-conscious refinement of hypotheses, models, and assumptions moving forward.

Our Key Research Questions

We sought to investigate the potential impacts of generative artificial intelligence (GenAI) LLM-based systems (in the form of coaching bots) on multiparty negotiation instruction. To explore such situations systematically, we structured our research around four primary questions:

1. Can students prepare more effectively for multiparty negotiations with the help of scenario-specific AI coaching bots?
2. Can we use back table player bots to simulate back table engagement prior to scheduled negotiations? What impact might this have on the outcome of multiparty negotiations (and on individual negotiators)?
3. With the help of coaching bots, can student preparation for and debriefing of assigned role-play simulations enhance negotiation outcomes and add to student skill development?
4. How can the use of AI negotiation coaching bots help students be more reflective and develop a more expansive and refined personal theory of practice (PTOP)?

We designed negotiation coaching bots to help students prepare and debrief privately so they can reflect on and learn from their negotiation experience in specific role-play simulations (Susskind et al. 2024). To deepen the coaching and preparation for multiparty negotiations, we added a step to what usually happens in a multiparty simulation, allowing each negotiator to engage with a bot representing the back table of each of the other parties with whom they are scheduled to negotiate. They could only talk with one back table representative at a time. This

In the following sections we first describe this additional backtable bot step and how it fits into our pedagogical process. We then describe our experimental research design, stressing why different kinds of bots are required, and detailing how we developed our bots with excerpts from our bot design prompts. Finally, we present our results with some illustrations of student-bot interactions and quotes from students describing their first-hand experience with a backtable coaching bot.

Before getting to the negotiation table, a good negotiator in a multiparty situation will attempt to gather as much information as possible about the other parties (Trif et al. 2023). In addition to consulting open sources or outsiders who have relevant experience with the parties who are the focus of attention, a negotiator may try to reach out “beyond the table” to have a conversation with someone’s back table. This is very hard to arrange on short notice. A back table contact, if you can reach them, may be part of another party’s organization, but may not necessarily be the decision-maker or know about the specific upcoming negotiation. The information they provide during this back table conversation (see example in Figure 6) may be quite partial or inaccurate.

Back Table Conversation



Our innovative idea was to add a preparation (coaching) step that makes it possible for each student to understand more about the other parties with whom they will be negotiating. After preparing normally using the confidential role instructions, the designated negotiation representative (negotiator) is provided an opportunity to hold a conversation with back table stakeholders for other parties. Negotiator (student) A reaches out to back table stakeholder (bot) from party B, and has a conversation (step 3 in Figure 7). This conversation can add to student A's understanding of Side B and may influence the rest of their preparations. They then can proceed to have similar conversations with the back tables of as many of the other parties as they want. After each conversation, back table stakeholder B (bot) summarizes in writing what was discussed, and sends their side's designated negotiation representative a copy of what they shared and what the bot says they learned about side A from the conversation (step 4 in Figure 7). When all these side conversations are completed, the parties then meet at the negotiation table (in class) to negotiate (step 5 in Figure 7).

Let's look at this process from the perspective of the student playing the role of the Regional Union's representative, one of six students who will be in-person at the table (Figure 8) in the upcoming class.

Figure 7
Example of Backtable Bot Implementation
(Student A – Backtable Bot B)

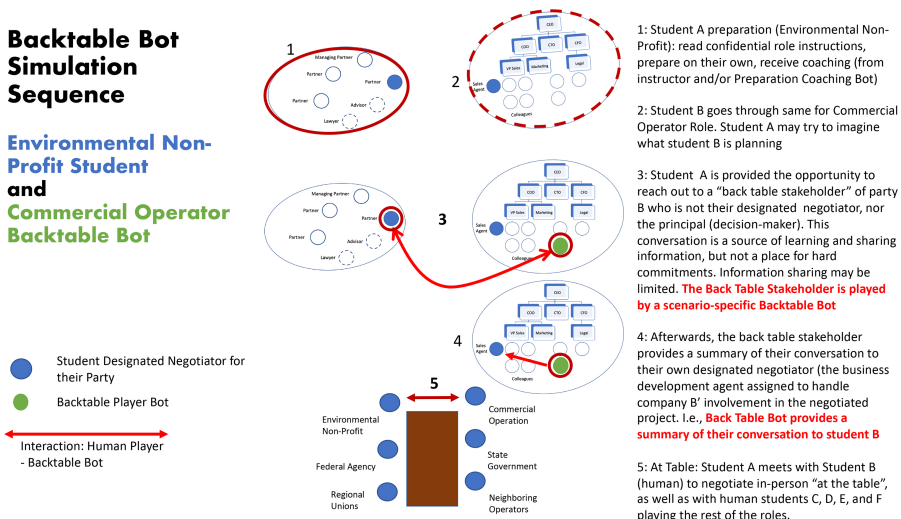
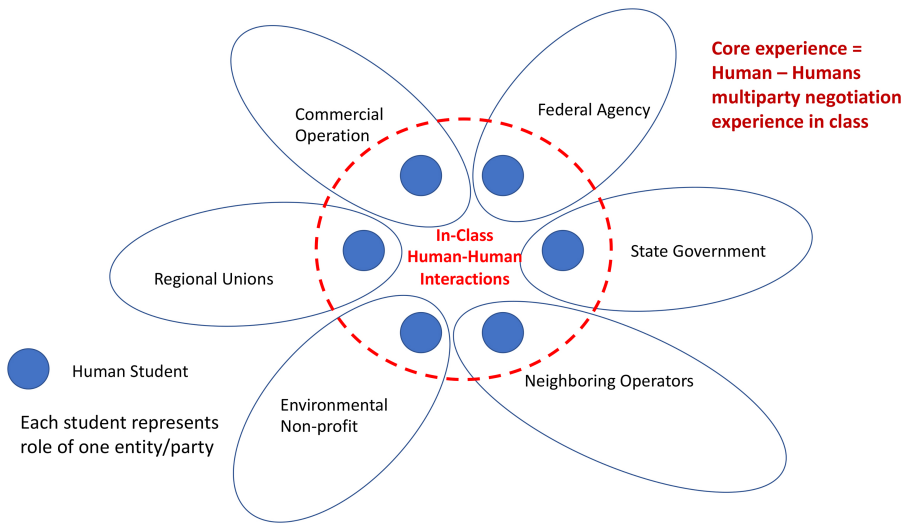


Figure 8
Illustration of a Six-party Simulation for Six Students
to Negotiate In Person



The teaching steps we traditionally follow are steps 1, 2, 3, 7, 8, 10, and 11 in [Figure 9](#). The new steps we introduced because of the coaching bots are step 4 (Preparation with Coaching Bot) and step 9 (Debrief with Coaching Bot). The new steps we introduced because of the backtable bots are step 5, in which the student is offered the opportunity to engage with each of the other parties' backtable bots ([Figure 10](#)); and step 6, in which the backtable bot provides—right before the assigned negotiation begins—a complete written summary of any and all backtable conversations that it had with any of the human players representing the other parties ([Figure 11](#)).

The benefits of using a bot for this back table conversation step are:

- 1) the conversation can be done asynchronously (each student moving at their own pace),
- 2) groups of six can be refashioned at the last minute on the learning platform (as a way of handling student absences),
- 3) none of the bot-human pre-negotiation commitments are binding, since these (by definition) do not involve the designated negotiation representatives, and the bots are designed not to make any absolute promises; and
- 4) the bot is designed not to disclose information it should not share.

Figure 9
Student's Learning Activity Sequence of Stages for an Assigned Simulation

Step	Student's Learning Activity Sequence of Stages for an Assigned Simulation
1	Read Instructions (General Instructions and Confidential Role Instructions)
2	Prepare on Your Own
3	(Submit Preparation Plan)
4	Prepare with Preparation Coaching bot (one time or many; various coaching styles)
5	Pre-Negotiate with Backtable Bots (with 1-5 of the other party back tables)
6	Receive Update from Own-role Backtable Bot (summary of 0-5 interactions)
7	Negotiate in Person
8	Reflect on Your Own
9	Debrief with Debriefing Coaching Bot
10	(Submit Written Reflection)
11	Debrief in Class

Figure 10
Union Party Student's Engagement with the Five Other Backtable Bots

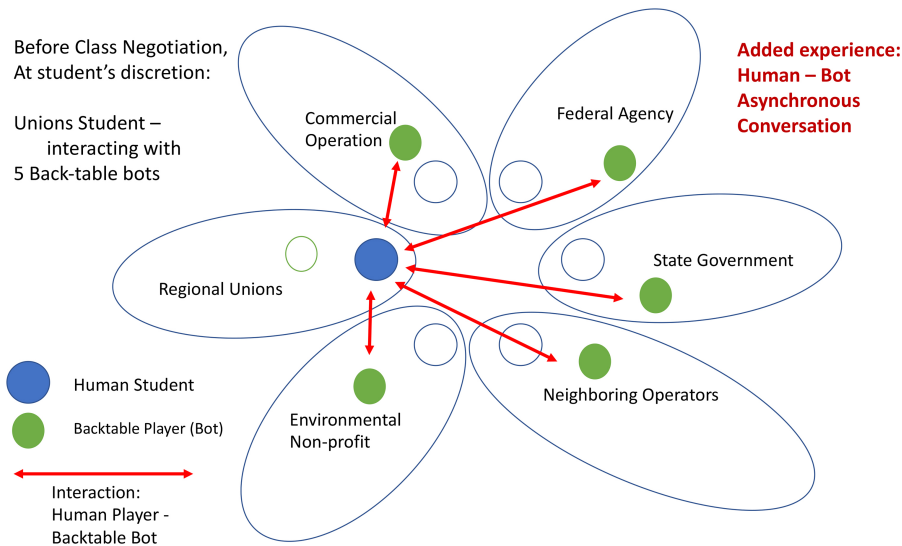
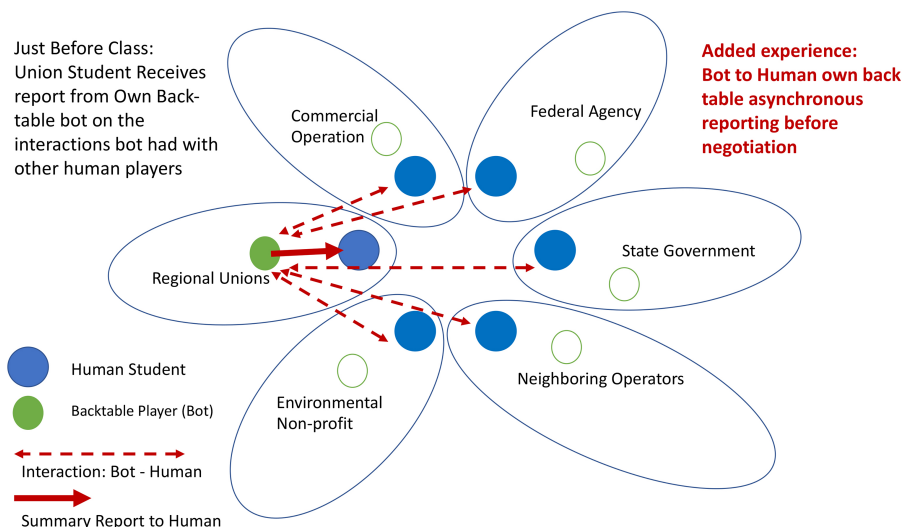


Figure 11
Union's Backtable bots' Engagements with Other Students,
and Summary Report



These benefits are confirmed by our students. One student commented:

I liked that I was able to listen and find more information about each of the other parties that would be there, in a way that was not as stressful because I knew that the people I was talking to (in the backtable) were not going to be at the negotiation itself, so there was less tension and expectation.

Another student said:

With the bots I feel like I heavily employed the mutual gains approach since I was able to understand everyone's motives and objectives via talking through the backtable.

Experimental Design—Participants, Setting, and Implementation

Our study was conducted during the spring semester of 2024. Participants included students enrolled for credit in our negotiation courses. Our sample size was limited to the 38 students taking our two graduate courses at MIT (with a limited number of student volunteers from other courses at MIT, to complete groups of 6 in certain negotiations). In addition, our study did not include a formal control group. There was no way to know before the semester began what the composition of

each of our classes would be. And, even if we knew, there would be no way to “match” the backgrounds of a control group for each class that we might have recruited. We were able to make comparisons, however, between our two classes, with one offered in the School of Engineering and the other in the School of Architecture and Planning (i.e., applied social science). And, in each of our classes, only two of our multiparty negotiation simulations involved coaching bots. So, we had some sense of how the student experience of working with the negotiation bots compared to the experience of working without the negotiation bots. We have many decades of running some of these same simulations in negotiation courses without bots. So, for the two role-plays for which we developed multiple kinds of bots, we had a very good sense of what the outcomes (and student reactions) would probably have been if negotiation coaching bots had not been used.

The changes we saw when the bots were added to two of the role-play simulations in our classes were very vivid. We collected a great deal of feedback from every student via surveys before, during, and after using each of the three kinds of negotiation coaching bots. Since we implemented three different coaching bots all at once in two role-play simulations, we are not able to separate out the impact of adding each type of bot (i.e., preparation coaching bot, debrief coaching bot, and backtable conversation bot). While the overall number of students involved limits the generalizability of our findings, the approach we chose, in our view, yielded very valuable preliminary insights into the impact of backtable bots on negotiation instruction.

Experiment Data Collection and Measures

Data were collected through a combination of bot interaction logs, student feedback surveys, and reflective journals. Specifically, we tracked:

- **Interaction Metrics:** We tracked the number of interactions each student had with the backtable bots, including queries made and responses received. Since the back table conversations were optional, it was useful to track how many students used this option, how many such interactions each student chose to initiate (each student could interact with one or more of the other five roles), and how much of their time each student invested in these conversations in order to learn more about the other parties by asking questions and exploring various factors that might impact the upcoming negotiation. Backtable bots were programmed to support their role’s front-line negotiator, and later to report the summary of their discussions to the student assigned to be that negotiator in the in-person class negotiation.

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- **Bot Execution Compliance:** We tracked whether the bot performed as was expected regarding sharing information and interfacing with the human students. For example, regarding confidentiality, we tracked instances where students attempted to elicit confidential information from the bots. In all observed cases, bots adhered to instructions not to disclose sensitive data, demonstrating robust compliance mechanisms.
 - **Qualitative Feedback:** Students provided qualitative feedback on their experiences with the backtable bots, highlighting perceived benefits and challenges.

We acknowledge our study's limitations, primarily our small sample size and the absence of a formal control group. Future research might aim to replicate our findings with a much larger, randomized sample of learners to enhance the robustness and generalizability of the results.

To ensure compliance with the role-confidentiality requirements of the simulation, as noted, backtable bots were programmed to withhold certain confidential information, which, if revealed, would disadvantage the front-line negotiators. Across 193 interactions, bots effectively maintained confidentiality, even when prompted to disclose restricted information. For example, when a student explicitly requested confidential details (i.e., what their front-line negotiator's sense of their BATNA was), the bot responded with a standardized refusal, thereby preventing information leakage.

Despite these safeguards, the use of LLMs presents challenges, such as hallucinations (Huang et al. 2023)—instances where bots generate inaccurate or fabricated information (Perkovic et al. 2024). In our context, hallucinations manifested as off-topic responses or the creation of fictitious negotiation issues. While creativity in responses can enhance negotiation training by simulating realistic and dynamic scenarios, it also risks crossing into fabrication, potentially misleading students. We carefully calibrated the bots' prompt designs to balance creativity with factual accuracy, employing iterative revisions and continuous monitoring to minimize hallucinations.

Additionally, LLMs can exhibit overly agreeable behaviors or struggle with complex reasoning tasks (Plaat et al. 2024). This could have undermined the authenticity of the negotiation simulations (An et al. 2024). To address these issues, we integrated specific prompt engineering techniques aimed at fostering balanced and reasoned interactions. This included setting clear guidelines for the bots' negotiation strategies and response patterns, thereby enhancing their reliability and effectiveness as coaching tools.

In summary, our research design attempted to meticulously integrate GenAI-driven backtable bots into multiparty negotiation instruction, addressing key research questions while acknowledging and mitigating inherent limitations. We elaborate on the outcomes of this design—providing evidence of the bots' impact on negotiation preparation and reflective practice—in the sections on building generative AI bots for multiparty negotiation and on the implications for the use of AI in teaching negotiations.

Context is Everything: Our Selected Role-Play Simulations

In negotiation, “context is everything.” We know this from decades of detailed case studies and related research reports on negotiations of all kinds—business deals, community disputes, and international diplomacy. That is, whatever approach a negotiator usually relies on must be modified or tailored to the specific situation or setting in which they find themselves (Schön 1984). They need to prepare by identifying the key considerations unique to the cultural context in which the negotiation takes place. While there are general negotiation assumptions and prescriptions that are almost always relevant, when and how to apply them, what adjustments to make, and when to emphasize different considerations, depends on each negotiator being able to draw on their own personal theory of practice (PTOP). That is, negotiators need to consider the unique cultural norms, organizational and institutional dynamics, and interpersonal expectations facing their negotiation partners and themselves in each negotiation. Knowing how to repurpose generally relevant tactics, strategies, and communication techniques as well as other PTOP elements can be the key to a successful negotiation practice.

In designing our negotiation coaching bots, we tried to incorporate the same theory and practice prescriptions we always present to our students during a whole semester course, as well as the ways in which these need to be adjusted so they match the specific context in which the student negotiators find themselves. We emphasize the importance of helping learners assess key aspects of culture, law, economics, and social dynamics that apply in particular contexts, and then formulate questions to help them make appropriate PTOP modifications.

To effectively evaluate the impact of backtable bots on multiparty negotiation instruction, we employed two distinct negotiation simulations: Harborco and Hydropower in Santaes (both available at <https://www.pon.harvard.edu/store/>). These cases were selected to encompass a range of multiparty negotiation dynamics and complexities, providing a robust framework for assessing the versatility and efficacy of GenAI-driven coaching tools.

In multiparty negotiations there are increased complexities of discussion process, group decision-making procedures, emerging coalitions, and shifting alternatives. When cross-cultural communication is also at the heart of a role-play simulation, all the negotiators must make sure they understand what the other role players are saying and that their own statements are being understood (Lee et al. 2013). When parties are working from different perspectives and cultural assumptions, miscommunications can easily get in the way of understanding and problem-solving. How to compensate for such differences and make appropriate adjustments in one's general approach to negotiation must be a part of each negotiator's PTOP. This includes knowing when to listen more carefully, managing one's cultural biases, changing the pace at which statements or ideas are exchanged, making sense of the ways in which seriousness and humor are being used, and understanding how one's counterparts probably relate to their back table.

Harborco is a six-party, multi-issue negotiation simulation involving representatives from a port developer, a labor union, an environmental coalition, other regional ports, the governor's office, and the federal department of coastal resources. The primary objective is to negotiate the building of a new deepwater port on the east coast of the United States. The structure of the negotiation process is three formal voting rounds, each requiring verification that the parties were voting in a manner that was consistent with their confidential scoring instructions. This setup emphasizes strategic coalition-building, issue prioritization, and formal decision-making processes, allowing students to engage in caucusing while conducting complex, scoreable negotiations that mirror real-world port development scenarios.

In contrast, Hydropower in Santaes ("Santaes") is a six-party, multi-issue mediated negotiation simulation set in the fictitious South American country of Santaes. Participants represent the leader of a local community, a hydropower plant developer, an environmental NGO, an Indigenous community, the mayor's office, and a mediator. The negotiation centers around a proposal to build a new hydropower plant. The discussions are divided into two phases: contract terms and decision-making procedures followed by mediation of the detailed terms of an agreement. Santaes emphasizes the importance of cross-cultural interactions, highlighting how cultural contexts and biases influence perception and, consequently, negotiation dynamics. Unlike Harborco, Santaes does not involve numerical scoring; instead, it requires participants to develop various packages and coalitions without reliance on numerical instructions, fostering a more subjective and fluid negotiation environment.

The similarities between the two cases are that they both involve six distinct parties with differing interests and priorities. Each party

represents multiple stakeholders (a company, a government entity, etc.) and so a backstory can easily be developed regarding a back table stakeholder for each role. Both cases require harnessing coalitions in order to influence the outcome and reach agreement.

But the differences between Harborco and Santaes also had critical implications for the design and implementation of negotiation coaching bots for each exercise. Harborco's structured, scoreable format necessitated bots that can effectively manage and verify numerical data, support coalition formation, and navigate formal voting strategies/procedures. In contrast, Santaes demands bots capable of understanding and adapting to cross-cultural assumptions and biases, facilitating mediated discussions, and handling the absence of numerical frameworks by promoting creative and flexible outcomes in response to the parties' confidential instructions. Additionally, Harborco has six parties with particular interests represented by their own personal scores, and some with hidden motivations regarding whether or not to build consensus. In contrast, the presence of an impartial mediator in Santaes requires the bots to support more nuanced information management and encourage balanced dialogue without overshadowing the mediator's role. These differences underscore the need for tailored prompt engineering and information control mechanisms to ensure that backtable bots could effectively support diverse negotiation scenarios, enhancing their utility across the two instructional contexts.

Building Generative AI Bots for Multiparty Negotiation

We resolved early to create bots unique to each role-play simulation with the key pedagogy of each simulation in mind (instead of attempting to build one generic platform for all simulations). One by-product of our work is a seven-step approach to designing GenAI tools to augment the use of role-play simulations to teach negotiation.

Seven Steps for Designing GenAI Tools

- *Step 1: Pedagogy Process – Choose carefully the case you want to implement. Decide how it will fit into the overall class syllabus, and be clear about your pedagogical objectives.*

Our strong motivation was to enhance human-to-human interaction, so we made sure that the backtable bot experience would inform the in-person negotiations, but not replace them. We chose simulations in which each role represented a party (i.e., a group of people or an organization) and thus we could claim that the engagement with the “back table entity” did not obligate the actual representative to say or

do anything in particular in the in-person negotiation. As noted earlier, when a party negotiates, there are usually many differences of opinion within the party itself (e.g., the organization or company) and internal negotiations “behind the table” between different groups and individuals. This “distance” between the back table entity and the designated negotiation representative (DNR) (i.e., the student who plays the role during the in-class negotiation session) enabled “plausible deniability” of anything that was said in the “pre-negotiation” back table interaction and the positions or interests the DNR presented at the table. In both Harborco and Santaes, each party represents an organization (such as the governor’s office or the union) or a group of people (such as the local residents or the Indigenous community). Therefore, the negotiators must consider important multiparty negotiation concepts such as understanding the interests of one’s various negotiating partners, agreeing on ground rules that will govern the parties’ interactions, building winning and/or blocking coalitions, and creating value through “packaging” or “trading across issues.”

- *Step 2: Pedagogy Objectives – Define specific tasks and patterns of engagement for each of the backtable bots.*

Each type of bot must reflect the instructor’s teaching objectives. Not only are there objectives specific to each case that is taught, but the learning objectives may vary from role to role within each case. While a preparation coaching bot will pose some similar questions to all the role players, some additional questions will be specific to each role. For example, in the Harborco simulation, all roles must consider who their allies and opponents might be, but each role has a different level of urgency regarding various possible agreements. One role is actually a spoiler—a party who wants there to be no agreement. One reason to engage in a pre-negotiation back table background discussion is to find out each party’s interests and priorities, and to figure out whether they might be an ally and a possible coalition partner.

In designing the preparation coaching bots,⁴ we prioritized (1) reinforcing each student’s understanding of the background case materials, (2) clarifying their motives and priorities in the role they have been assigned, and (3) anticipating contingent responses to a range of tactics and strategies that the other parties might use. “Putting yourself in the shoes of the other party” is a very important component of negotiation preparation. In multiparty negotiation, this is a point at which many students (and most negotiators in general) fall short (i.e., they do not devote enough time to understanding the perspective of each of the other parties). This is where the added step of communicating with back

table representatives of the other parties is extremely helpful; it “forces” the student to engage with the “someone” who knows what their negotiating partners are thinking.

We also gave a secondary objective to the backtable bot: to find out as much as possible about the student’s confidential instructions. This meant that the bot (e.g., the back table member of Party B) was waiting to ask questions of the student (Party A) about their understanding of their party’s (A’s) interests, priorities, and strategy. The bot is told to summarize and store whatever information it finds out, so that a report could be fed to its assigned student before the negotiation begins. (So backtable bot B would have a report that includes summaries of the conversations it held with students A, C, D, E, and/or F and the report eventually is made available on the learning platform to the student playing role B.)⁵

This “update report about back table conversations” was programmed to be given to each student right before the scheduled negotiation. This represents a new pedagogical objective: learning to deal with “last minute” updates from one’s back table that include new information.

We selected parts of each role’s confidential instructions that were incorporated into the questions and responses that the backtable bots were allowed to disclose about their interactions with different players. We instructed the backtable bots as to what information they could not share (for example, their role’s scoring structure). We also instructed them that they could not make hard commitments regarding coalitional support or future voting.

- *Step 3: User Experience Design – Describe in detail how you intend for students to interact with and learn from the coaching bots.*

Based on years of teaching negotiation, we tried to imagine what this back table engagement might sound like. This was basically a two-party negotiation, but one where reaching a deal is not the desired outcome. Rather, the goal is to obtain only a preview of key parts of the negotiation—finding out interests, options, measures of success (i.e., criteria), alternatives, and priorities. Given the current technology, this interaction took the form of back-and-forth text messages between the student and the bot. We considered who should start “speaking” and what the introductory statements on both sides might sound like. We drafted possible exchanges, taking account of how best to handle breaks or interruptions. We concluded by thinking about the ending of each session: what summary items or accounts ought to be offered (e.g., main takeaways), what questions should be asked before ending (about

the session itself), and how to verify with the student that they wished to end the session. We drafted imagined scripts for every backtable bot pair for each role in each case.

We realized that just as each human coach has a personality, we needed to assign each bot a communication style or personality.⁶ We decided to set one fixed personality for each backtable bot. (We did this so that all five other parties spoke with the same back table player and that player had the same personality no matter which party it spoke to. In the future, other personalities might be possible.) Students reported forming stronger connections to AI bots that exhibited certain human-like interactions that made them more comfortable. One student captured the resulting ease of communication by saying, “The interaction was well articulated and felt very natural. That was useful to me because I registered right away what the important things were, so it was very effective for me.”

One key personality trait that we focused on for the backtable bots was reciprocity. We instructed the bots to be reciprocal in their sharing of information: share a little bit of information and expect to receive some information in return. If the student was collaborative and sharing information, then the bot would do the same, and continue to share information until it reached its assigned limit (“I am sorry, but I cannot answer that question”). If a bot senses that the student is only asking questions and not sharing information, then the bot was instructed to not share any more information until the student reciprocates. If the student was not collaborative, the bot would shut down the conversation.

- *Step 4: Integration Planning – Plan your integration and testing based on a thorough understanding of the requirements and constraints of the current technologies.*

An important aspect of integrating these AI tools into our teaching involved aligning them with the operational and technological constraints of the advanced online learning platform we had selected. We needed to fully understand the strengths and weaknesses of current technologies, the learning platform structure and its limitations, the state-of-the-art of current GenAI tools, and the available integration interfaces. We had to be willing to revise our earlier choices and aspirations. Since these backtable bot conversations are essentially similar to the beginning of a two-party negotiation (without trying to close a deal), we can rely on advancements in two-party negotiation bots. During this past semester, we had to deal with the learning platform’s limitations and integration capabilities, such as restrictions on the length of messages and interactive sessions, and the amount of material that can be

used as input to a prompt. At the time of this writing, we are already making progress on switching to more advanced tools that might enable voice negotiation or even video negotiation (human to avatar).

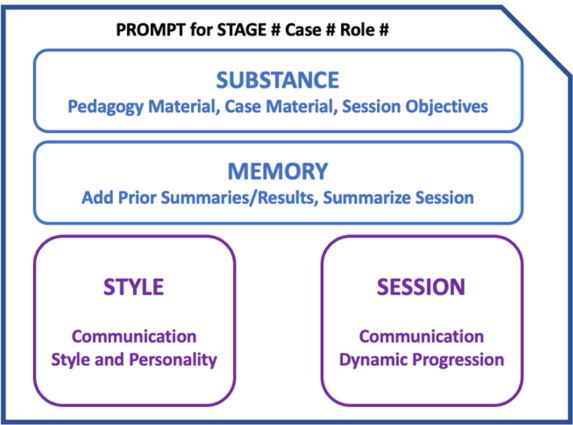
- *Step 5: Prompt Design – Develop and test each coaching bot’s AI prompt in full detail by using direct engagement with a selected LLM. Commit to continuous revisions.*

Large language models (LLMs), which underpin our coaching bots, are driven primarily by sophisticated prompt engineering. Prompt engineering has evolved substantially, with numerous studies (e.g., [White et al. 2023](#)) illuminating the subtleties of prompt structure and its important role in eliciting effective AI responses. We leveraged our prompt developers’ extensive experience with ChatGPT to craft prompts that not only served our educational purposes but also ensured the integrity of the saved interactions. We used ChatGPT-4 Turbo.

Prompt design is at the core of creating negotiating coaching bots of all kinds, including backtable bots.⁷ We used four distinct aspects of prompt design, focusing on 1) the substance of the conversation, 2) the style of the conversation, 3) communication session dynamics, and 4) the bot’s “memory.” [Figure 12](#) shows the four aspects of bot development.

Each role-role conversation in the two negotiation scenarios was assigned a backtable bot design. Separation prevented cross-role information leakage. The backtable bots’ “memory” incorporated the summary

Figure 12
Four Aspects of Prompt Design for the Negotiation Coaching Bot



of each student-to-bot conversation into a later “update report about the back table conversations” for each student role player. To illustrate the power of the bot’s memory, consider the following scenario that deals with the issue of last-minute changes to student group assignments.⁸ Assume that as instructors, we assign the following students to roles A–F in a six-party exercise: Adam, Beatrice, Charlie, Dana, Evan, and Farid. Let’s assume all students have taken advantage of the opportunity to conduct back table conversations with all of the other five roles’ back tables. Now, let’s assume an hour before class we learn that Adam will not be able to make it to the in-person exercise. We assign a different student who also had prepared for role A on the learning platform, Ali, to step in. We update the group on the learning platform before releasing the last minute update report about the back table conversations. Now when the report is released, students B through F see in their report the summary of Ali’s conversation with their own backtable bot (and not Adam’s conversations).

- *Step 6: Integration Debugging – Integrate your tested prompts into the learning system’s interface and perform integration testing.*

After rigorous direct testing of each prompt with GPT-4 Turbo to confirm the effectiveness and security of each set of bot interactions, we integrated all the prompt sequences into the learning platform (iDG). Our integration ensured that we could immediately access summaries of each student’s interactions. For testing, we used staff and volunteers to ensure each role in the group exercise worked correctly for all stages of the exercise.

- *Step 7: Refinement – Development team evaluates and adapts its earlier choices.*

The development of our negotiation coaching bots was inherently iterative, requiring numerous rounds of testing and refinement by the full teaching team.⁹ After ensuring basic functional integrity, we continued fine-tuning each of the prior step results, from pedagogy to prompts and integration.

Each bot session prompt was developed by a designated prompt developer. The developer wrote the prompt and tried it out in dialogue with ChatGPT-4 in a stand-alone session. That session was then reviewed by the designer and an experienced instructor, who often suggested corrective actions. The developer could then repeat the process until each prompt produced the desired effect. The key to good prompt design is being very precise. It is a craft that one gets better at with experience.

Then, the developer must “stress test” the prompt to see how it handles various user behaviors, and, where needed, continue to refine the prompt. Once the prompt is deemed acceptable, it is then loaded onto the learning platform in the right place in the flow. Then, the developer must test how the various prompts work together and make any minor tweaks specific to this “final use” context.

Students accessed their negotiation coaching bots through the learning platform, and therefore could navigate this platform themselves to access a particular simulation, and within that simulation, view their instructions, input any agreements made along the way, note any voting results, etc. We incorporated our negotiation coaching bots at key moments in each simulation, to allow students to interact with the bots before and after they completed their in-person negotiation.

AI Model Selection and Configuration

At the inception of this project, we opted to utilize the latest available GPT model from OpenAI, specifically GPT-4-Turbo-Preview.¹⁰ This choice was informed by the model’s advanced capabilities in natural language understanding and generation, which are critical for simulating realistic negotiation interactions. We configured the model using default parameters, setting the temperature between 0.5 and 1 on a scale of 0 to 2. This range was selected to balance creativity and coherence in the bots’ responses, ensuring that interactions remained both engaging and contextually appropriate.

Prompt Engineering and Cognitive Strategies

A pivotal aspect of our work involved sophisticated prompt engineering techniques to enhance the accuracy and relevance of the bots’ outputs. We employed Chain-of-Thought (CoT) prompting, a strategy shown to significantly improve the reasoning and problem-solving capabilities of large language models (Wei et al. 2022). CoT prompting involves guiding the model through a step-by-step reasoning process, thereby enhancing the logical consistency of its responses. In our case, we built much of this step-by-step reasoning into the conversation structure and the communication style of our negotiation coaching bots. Thus, when a student queries a bot about potential coalition strategies, the prompt guides the model to consider first the possible options before selecting the most viable one based on the negotiation context, and any additional restrictions that the instructors feel are necessary. For example, in the back table interaction scenario, as part of the CoT the bot would come up with a series of responses, filtered by the instructor-imposed limitations on what can be disclosed to which party.

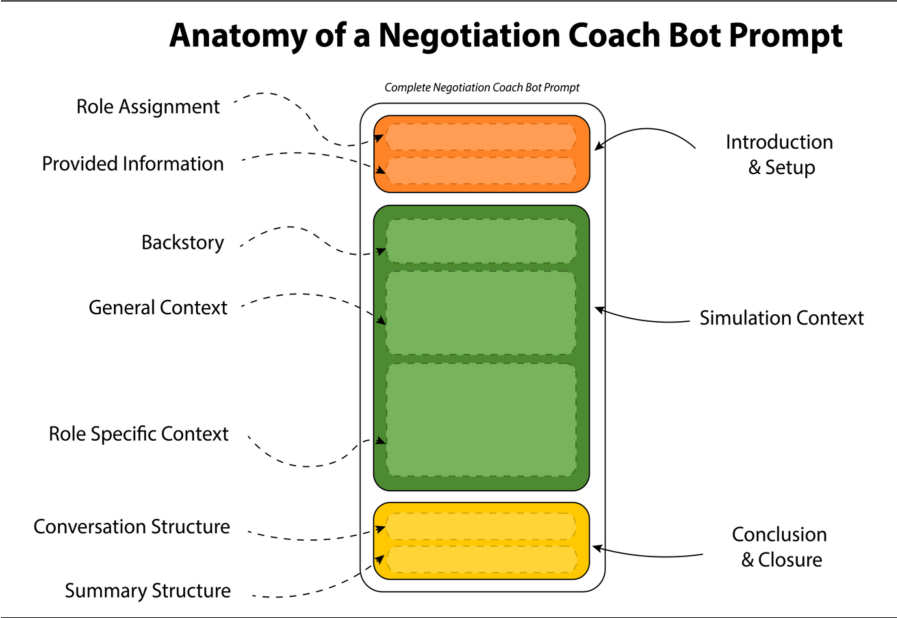
Additionally, we integrated *few-shot learning* (Song et al. 2023) alongside zero-shot CoT prompting to refine further the bots’ conversational

behavior. Few-shot approaches provide the model with a few example interactions, enabling it to understand better the desired response patterns and context-specific nuances. For instance, in our conversation structure we included explicit examples of how a student might try to get more information out of the backtable bot without divulging much on their end, and how the bot is meant to respond. We also included conversation starting phrases while accounting for each bot's assigned personality and communication style. This combination allowed us to craft complex prompts that could dynamically adjust to various conversation threads, ensuring that each interaction remained tailored to the individual student's needs.

To understand how this came together see [Figure 13](#), which illustrates the finalized structure of our prompt; the components are listed on the left side of the figure and detailed below.

- *Role Assignment:* The coach role is laid out, clearly identifying the conversation participants and noting the overall goal of the conversation.
- *Provided Information:* Lists the information provided in the rest of the prompt to facilitate the role, behaviors, and conversation as outlined.

Figure 13
Anatomy of a Negotiation Coach Bot Prompt



-
- *Backstory*: Provides background, personality overview, and communication style (back table role added here to differentiate from negotiator role).
 - *General Context*: This is the generic simulation context as provided in the existing instructional materials. These give context on what the negotiation is about, who the interested parties are, and what dynamics are at play.
 - *Role-Specific Context*: These are the instructions specific to the role that the coaching bot will need to play. This will include a list of priorities and the information-handling protocol (i.e., what can be shared, might be shared, and must never be shared).
 - *Conversation Structure*: This is a strict structure to follow for the conversation that guides the interaction and sets rules for necessary elements/interactions.
 - *Summary Structure*: This is a structure to guide what information is captured to represent the interaction and what is passed back to the learning environment.

Sample Prompt Excerpts

The back table coach is designed to simulate interactions with a counterpart's back table, providing a realistic, role-driven dialogue that balances information sharing with strategic inquiry. By integrating the structured conversation framework, defined information handling rules, and personality-driven engagement, the back table coach offers students an immersive negotiation preparation experience, providing realism, trust building, and alignment with negotiation objectives. As one can see from the excerpts provided below, prompt design is an art in precise English writing, written as instructions in the second person. Thus, prompt design does not require any computer coding skills.

An example of a role assignment design prompt excerpt for the Harborco Union back table coach reads:

I am a student preparing for the Harborco negotiation exercise in my multistakeholder negotiations class. I am playing the role of <%player_role%> negotiating with the 5 other stakeholders in a negotiation involving a dispute over the construction and operation of a deep-water port off the coast of Seaborne. Some preliminary planning and design work has already been done, but it cannot proceed without a license issued by the Federal Licensing Agency (FLA).

And later...

You are Michael, a trusted advisor to the Union's lead negotiator, playing a crucial role in developing the organization's

bargaining strategy and providing critical support throughout the negotiation process. I am reaching out to you in my role as a <%player_role%>, and based on our prior interactions and your reputation within the local labor movement, to have a conversation in the hopes that I might gain some valuable insights that would help me better understand the Union's priorities and approach regarding Harborco's proposed deep-water port project.

An example of a provided information design prompt excerpt for the Harborco Union back table coach reads:

To facilitate your role as a Michael, I am providing you with the following: Backstory – information provided to you as a backstory to your conversation with me, and guidance on your personality and communication style for our conversation..... Keep in mind that I am playing the role of <%player_role%>, so any reference to <%player_role%> is a reference to me or the role that I will be playing in our conversation....."

The back table coach's design objectives include:

1. Realistic role simulation: Create an engaging and personality-driven character with defined traits and backstory.
2. Strategic information handling: Implement rules for sharing or withholding information based on the student's role and relationship with the bot character.
3. Structured interaction framework: Ensure conversations follow a logical flow to achieve learning objectives while maintaining authenticity.
4. Role immersion: Provide role-specific prompts to contextualize the student's approach to the bot and inform the bot's responses.
5. Trust building and relationship management: Encourage students to practice trust building and nuanced dialogue with a focus on mutual understanding.

The following is an example excerpt for backstory design prompt:

This is the backstory section with information provided to you, Michael, as a backstory to your conversation with me.

Background: Born and raised in a working-class neighborhood in Seaborne, Michael has witnessed firsthand the impact of economic fluctuations on the lives of hard-working families.... Michael is passionate, principled, and deeply committed to the cause of economic justice for working people. He approaches his role as an advisor with a keen sense of

responsibility, driven by the belief that the Union's success in these negotiations will have far-reaching implications for the lives of its members and the broader community. Michael is known for his ability to analyze complex situations, anticipate challenges, and devise creative solutions that balance the needs of workers with the realities of the negotiating table.

And here is an example excerpt for information handling design prompt, where the backtable bot should respond to user inquiries while strategically asking its own questions to gather insights:

This is the information handling section which provides a collection of the information you can share and cannot share during our conversation based on the role that I am playing.

Use the information provided and guidelines on what can and can't be shared above to converse with me. If I ask you a question about the negotiation with your superior and what they might want, reply to the question and then follow up with an inquiry of your own about the information you would like to gather about what I want in my role as <%player_role%>.

Finally, regarding closing example design prompt excerpts, the bot should ensure alignment by summarizing key points and encouraging reflection:

After addressing any additional questions or concluding any additional inquiries, you will mention that you hope our conversation has been helpful and ask whether I think that we are aligned on goals or objectives. Remember to maintain your personality, character, and communication style in choosing how to ask me this question.

Findings During the Process of Developing the Bots

A primary challenge in implementing LLM-driven educational systems, particularly for backtable bots, is the prevention of information leakage. Unlike preparation and debriefing coaches, which can reference a broad range of simulation contexts, backtable bots must strictly limit their communication to predefined information. To address this, we established rigid information controls embedded within the prompt structure. Instead of deploying a single backtable bot with a universal prompt, we created six distinct bot instances, each corresponding to a specific role within the negotiation simulation. This segmentation ensured that each bot had access only to the information pertinent to its designated role, preventing inadvertent disclosure of sensitive details that the designated negotiator representing their party may have. This was especially crucial in Harborco, where one of the parties may be

acting as a spoiler to the negotiation but we do not want the back table stakeholder to “know” this so that they cannot inadvertently disclose it. The efficacy of these information controls was rigorously tested during the simulations across 193 distinct backtable bot interactions, each comprising multiple conversational exchanges. Throughout this extensive “in-use” testing, the bots consistently adhered to their predefined constraints, maintaining role-specific information boundaries. Notably, students were explicitly challenged to “pressure test” the bots and report any anomalies, yet no instances of unintended information biases were reported. This robustness was further corroborated through a preliminary analysis of submitted responses conducted by teaching assistants. Moreover, more structured—albeit informal—student interviews and targeted questions aimed at uncovering potential discrepancies in bot behavior yielded no contradictory findings. The only anomalies we found were: two wording differences (headers included in the first message text generated by the bot); and an unexplained early termination of a back table conversation by the backtable bot. In these anomalous instances there was no notable impact on the learning experience, nor on other parts of the conversation. At this point, we could not find the root cause of those anomalies within the complex system. Though a more detailed analysis (or a more extensive study) might reveal more nuanced disclosures, our multifaceted validation approach substantiated the effectiveness of our information control mechanisms in maintaining the integrity of role-specific knowledge boundaries within the LLM-driven negotiation simulation.

To foster more natural and effective interactions, we imbued each backtable bot with a distinct personality profile. This involved crafting detailed personality descriptions that anchored the bot’s behavior, making interactions more engaging and realistic. By explicitly referencing these personality traits throughout the prompt, we ensured that the bots maintained consistent behavior patterns aligned with their assigned roles. For instance, a backtable bot representing a union leader was programmed to exhibit assertiveness and a focus on worker welfare, while another bot representing management might prioritize cost efficiency and strategic alliances.

Given that LLMs can sometimes be overly agreeable or struggle with complex reasoning (Bender et al. 2021), we implemented additional prompt engineering techniques to mitigate these tendencies. This included setting clear guidelines for negotiation strategies and response patterns, thereby fostering balanced and reasoned interactions. For example, bots were instructed to present counterarguments and maintain firm stances on key issues, preventing them from conceding too readily and ensuring that students encountered realistic negotiation dynamics.

To ensure that conversations remained goal oriented and aligned with the negotiation objectives, we provided the bots with a structured conversation framework. This framework guided the interaction flow, outlining permissible topics and delineating the manner in which the bots should steer discussions. By enforcing a clear structure, we helped the bots prioritize critical negotiation elements and maintain focus on strategic objectives. Only a few resulting interactions included off-topic responses or unintentional shifts in discussion focus (as might happen in real life), and even those were slight and did not impact the student experience. We also encountered some instances of prompt injection, where students inadvertently submitted empty or unintelligible inputs, resulting in nonsensical or default bot responses. While these scenarios did not lead to any unauthorized information disclosure, they underscored the necessity for robust input validation mechanisms. To address this, future iterations will incorporate more stringent data validation protocols to filter and sanitize user inputs preemptively, thereby reducing the likelihood of successful prompt injection attacks and enhancing the overall security of the system. In fact, an error resulting from LLM hallucinations—instances where the model generates inaccurate language or fabricated information—had a less detrimental impact than initially anticipated. A backtable bot for the Union role erroneously promised agreement to a proposal, and it led to realistic student frustration upon encountering disagreement at the negotiation table. But such a scenario, while arising from model inaccuracies, mirrored real-world challenges that negotiators must navigate and offered a “teaching moment.”

Experimental Results: Student Engagement and Test Negotiations

The participants in our experiments were enrolled in “Multistakeholder Negotiation for Technical Experts” at the MIT School of Engineering, taught by Samuel Dinnar, and “Negotiation and Dispute Resolution in the Public Sector” in the MIT Department of Urban Studies and Planning, taught by Lawrence Susskind. In total, we had five groups for the experimental runs of the two games.¹¹

Here are the steps taken by both the students and the instructors to get ready for, and following, the in-class experiments:

- *A – Self-assessment baseline survey:* The self-assessment baseline survey¹² measured levels of negotiation skills among participants before they engaged with the AI coaching bots.
- *B – Role assignment and individual student preparation:* Students were assigned roles and given access to each simulation (one at

a time) via the learning platform. This is how they accessed their instructions and their custom coaching bots. Traditionally, after reading their written instructions and preparing on their own, many instructors require that students submit a written preparation assignment describing their strategy, fill out a negotiation preparation sheet, or answer a pre-negotiation survey. With our revised process, this can be replaced by the pre-negotiation bots' steps C and D.¹³

- *C – Pre-negotiation—individual preparation coaching session:* The pre-negotiation preparation coaching bot for each role had all the general information about the case, role-specific information, instructions on how the coach should interact with the learner, and instructions on how to summarize the conversation at the end of each coaching session (step 4 from [Figure 9](#)).
- *D – Pre-negotiation—back table interactions:* The back table coaching bot offered, in a time-limited way, an opportunity for each role player to have a brief conversation to learn more about the other roles' interests and intents. Each student was offered an opportunity to interact with the other roles' back tables. The simulation experience in class would not be impacted if a student was not able to, or decided not to, have any such conversations, or if a student decided to only have a few such conversations. As described above, the final stage of this step was to provide a summary to each student of all the conversations that the other students in their group had with their own back tables (steps 5 and 6 from [Figure 9](#)).
- *E – In-person negotiation:* After students completed the required individual negotiation preparation, they proceeded to have the multiparty in-person face-to-face negotiation in class.¹⁴ One student was assigned each role.¹⁵
- *F – Post-negotiation debrief—individual debrief coach:* Shortly after the negotiation ended, the post-negotiation individual debrief coach provided an opportunity for each student to reflect on their performance and the results of their negotiation (given how they described their negotiation strategy and objectives during their earlier preparation session), including how their actual experience differed from their preparations and their discussions with the various backtable bots (step 9 from [Figure 9](#)). The individual debriefing coaching bot prepared a summary of the conversation at the end of the individual debriefing for students to use in preparing their assigned written reflections. (Students were still asked to submit a written reflection as a homework assignment.) The students' level of engagement with this debriefing bot was uniformly high.
- *G – Instructor summaries and in-class debrief:* Summaries of each student's level of engagement, as well as conversation summaries, are

available on the system along with summaries of each test group's negotiation results. These were used by the instructor to structure the in-person in-class debrief.¹⁶

- *H – Self-assessment post-class:* The post-negotiation assessment survey aimed to measure the effectiveness of AI-assisted tools in enhancing negotiation skills (in light of how participants completed the exercise). This survey was structured to compare against the baseline established pre-negotiation.

Implications for the Use of AI in Teaching Negotiations

Our experience suggests that customized coaching bots can reinforce key learning points in negotiation courses. We are not convinced that building generalized AI-assisted coaching bots to be used with any and all role-play simulations without adjustments is currently a worthwhile endeavor.

Results from the Role-Plays

In the Harborco scenario, which involves numerical scoring metrics aimed at optimizing value creation, students engaged actively with the AI coaches. The negotiation coaching bots helped students leverage their understanding of each player's priorities to guide them in implementing strategies aimed at shaping voting outcomes, building coalitions, and using caucusing effectively.

Feedback from our students was overwhelmingly positive. To evaluate students' gains, we chose a 7-point Likert scale based on its reliability, validity, and widespread use in psychology and social science research to collect participant perception data (Kusmaryono et al. 2022). For example, in comparing pre-negotiation reports and post-negotiation reports regarding the use of negotiation strategies to discover other parties' interests, students indicated an improvement from an average (mean) and median of Moderately Confident (4.60/4) to Confident (4.94/5) on a 7-point Likert scale. Their reported ability to deal with strong emotions went from a mean of Moderately Confident (4) to Confident (5), and their ability to deal with bad actors—a key learning objective in the Harborco simulation—improved from being insecure (mean of 2, or average of somewhat insecure 3) to moderately confident (4). See Figure 14.

When asked about the ways in which the bots were most helpful, our students reported that the combination of the coaching bots and backtable bots was most helpful:¹⁷

- 82% felt better prepared in advance of the negotiation.
- 77% experienced improvement in their ability to discover others' interests.
- 59% found it easier to express their positions or priorities.

Figure 14
Sample Student Survey Results: Perceptions Before and After
Experiential Learning

Survey Results for 3 Notable Categories						
	Discover Others' Interests		Deal With Strong Emotions		Deal With Bad Actors	
Average	4.60	4.94	4.10	4.53	2.85	3.82
Median	4- Moderately Confident	5- Confident	4- Moderately Confident	5- Confident	2- Insecure	4- Moderately Confident
Standard Deviation	1.27	0.97	1.62	1.01	1.50	1.19
	Before	After	Before	After	Before	After

Likert Scale Used to Collects Students Perception of Their Own Capabilities (1-7) Before and After Exercise

1- Very Insecure	2 - Insecure	3- Somewhat Insecure	4 - Moderately Confident	5 - Confident	6 - Very Confident	7 - Extremely Confident
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Students appreciated the backtable bot experience, and many of them commented that it added to their ability to understand the interests and priorities of each of the other parties. For example, one student mentioned:

I was able to ask a lot of questions of the backtable bots which I feel was very beneficial and supportive of the outcome that our group arrived at. The bots helped me consider different conditions that I may have to bring up during the actual negotiation process.

Many students also mentioned that interacting with the bots forced them to think hard about their initial alignment with each of the other parties and their strategy for building a winning coalition. One student noted:

The most useful thing was the fact that I was able to go back to a different (back table) party after speaking with someone else. I could ask follow-up questions based off of what I learned from (the other back table interactions), and that facilitated a lot more preparation about which packages each party might be able to accept.

The additional benefit of our LLM-based bots in these negotiation simulations is their ability to actually reach out and have a meaningful conversation about the upcoming negotiation, as captured by another of our students during this experiment, who noted:

I think it's very useful to have the back table conversations because I had a lot of random questions to ask each of the

parties and yet I was never turned down. This allowed me to expand (my strategy) and think more about what direction I should take in the negotiation.

Naturally, this may not be the students' experience in the real world (or in a simulation created with a different prompt design).

After experimenting with both exercises in class (rather than only the Harborco), we again surveyed the students about their reactions. 22 students (of 24) answered the surveys and reported that AI-assisted coaching significantly improved their ability to draw lessons from their own experience. A remarkable 90% of our students valued the backtable bots for their helpfulness.¹⁸

As reflected in the survey, the students noted that their experience using the bots was especially helpful in boosting their confidence, preparedness, and understanding of other parties' interests, and in helping them to be more thoughtful in formulating their negotiation strategies. The backtable bot conversations helped them deepen their understanding of the differing perspectives of each party and revise some of their earlier assumptions,¹⁹ with one student capturing the value of this interaction by stating:

I could start putting in my interests and seeing where they would align with some of the other groups. I could approach each of the back table parties that I was concerned about or interested in forming a coalition with, and prompt the back table with some questions, or start laying the groundwork for compromise.

Another student noted:

Talking to the back table helped me understand the moves other parties might employ, as well as their positions and values relative to my own. I was less surprised during the negotiation as a result.

Students commented on how holding the backtable bot conversations, and then receiving a summary of the other parties' conversations with their own backtable bot, presented a more realistic multi-session feel to the simulation (which was originally designed as a one session in-person interaction) by adding the pre-negotiation steps. Students also appreciated the added challenge of having to adapt to the new information received just prior to starting the scheduled negotiation. In addition, they were surprised how interactions with their debriefing coaching bots clarified the differences between what they learned from the pre-negotiation back table conversations and the actions taken by the real-life student counterparts. Alongside these advantages, some students

acknowledged several challenges in translating coaching bot interactions into successful outcomes during their at-the-table negotiations.

To quantify student engagement with the AI-driven backtable bots, we employed a systematic approach to data analysis. We examined the conversations between students and backtable bots, counting the number of messages in each exchange. An exchange was defined as a conversation containing at least one message and one response (within a reasonable active window);²⁰ interactions failing to meet this criterion were excluded from the analysis. We also measured the duration of each conversation session.

We further parsed the chat data to determine which of the five available backtable bot instances each student interacted with. This granular analysis allowed us to disaggregate the data on interaction time and message frequency for each specific bot instance. This approach provided insights into not only the overall engagement with the AI system but also potential variations in student interaction patterns across different bot roles within the negotiation simulation.

This methodology enabled us to construct a comprehensive picture of student engagement with the backtable bots, laying the groundwork for more detailed analysis of the effectiveness and implications of AI integration in negotiation education.

Overall, the coaching and backtable bots were a valuable complement to first-hand (experiential) practice. Students increased their investment of personal time in negotiation preparation (versus prior class negotiations in simulations with no bots) through their interactions with the coaching and backtable bots. The vast majority of students (94.7% in the second exercise) took advantage of the opportunity to engage in back table conversations,²¹ with most indicating that it added a lot to what they learned.

For a single conversation with one backtable bot, students who chose to engage spent an average of 9 minutes per bot (with a standard deviation 7 minutes), exchanging an average of 11 messages per bot. One student sent as many as 43 messages back and forth with one of the bots over an 18-minute period (that student had a total of 147 back table message exchanges with the various backtable bots in the specific single negotiation simulation).

By the second exercise, students were spending an average of 8.5 minutes per back table conversation and engaging with approximately four back tables, resulting in an average total time of about 34 minutes spent on preparation by considering their negotiation counterparts' perspectives. It is worth noting that individual engagement varied considerably, which is to be expected for students with differing abilities to engage in additional preparation.

As discussed earlier, we used two similarly complex 6-player negotiation simulations, and the students were offered similar backtable bot opportunities for each game. To illustrate the evolution of engagement from the first to the second simulation exercise, Figure 15 presents aggregate game metrics for students who completed both exercises, and highlights substantial increases in engagement time, messages sent, and the number of backtable bots engaged. This trend corroborates the value students found in the backtable bots, as indicated in their interviews.

One of the most important outcomes was the impact of backtable bots on students' personal theory of practice (PTOP). The majority of students reported that interactions with the bots prompted them to rethink and refine their negotiation strategies and theoretical approaches. For example, one student who participated in the Hydropower in Santaes simulation exercise explained in their reflection:

This negotiation was where I felt that I most effectively communicated my priorities, by focusing on environmental assessment, job creation, and water level management. I identified all of these priorities with the help of the pre-negotiation AI bots.

A second student wrote:

I was able to talk with some of the backtable bots to see if I could see what their priorities were or at least determine their BATNAs. I was able to figure out where my thoughts aligned with theirs and how I could possibly come up with value creating moves that would benefit the entire group.

Figure 15
Summary of Metrics in Student Engagement with Backtable Bots

Metric	Exercise 1	Exercise 2	Change
Average Total Backtable Engagement Time (mins per student)	28.6	33.8	18%
Average Backtable Bots Engaged (per student, out of five)	2.7	3.7	39%
Average Message Exchanges per Bot Conversation	10.4	11.2	8%
Student Engagement (% of students in class)	96%	96%	No Change

This evolution in PTO aligns with our success criteria, emphasizing the transformative potential of GenAI tools to foster deeper, more reflective learning experiences.

Texts of Student Interactions with the Backable Bots

Figure 16 includes several screenshots of the interactions in the Santaes simulation between a student (Dana) playing the role of the Junta de Vecinos party (messages on the right side) and the backtable bot from the mayor's party (with the backstory of the backtable bot being that its name is Martin, the Chief of Staff for the mayor). Bot messages are left justified, and the bot starts the conversation.

Finally, Figure 17 includes a screenshot of how the Ortega company backtable bot closed and summarized their back table conversation with the same student (Dana, playing the role of the Junta de Vecinos party). This summary was then held in "memory" by the learning platform to be used in the following ways:

- To be shared with the student who will be playing the designated negotiator role for Ortega, so that they know what was shared with Dana when they meet at the table, and can plan accordingly.
- To be shared with the instructor, to know how Dana used the backtable bot feature.
- To be shared with Dana and Dana's debrief coaching bot later, after the negotiation, when debriefing Dana's plans and strategy going into the negotiation versus the results at the table and her thoughts afterward.

Reactions from Professional Colleagues

After experimenting with the bot-assisted role-plays and debriefing our students and teaching assistants, we had an opportunity to present our preliminary findings to a group of 30 experienced negotiation professors and instructors gathered at a pedagogy dinner hosted by the Program on Negotiation at Harvard Law School. This semi-annual Pedagogy at PON²² session helped us refine our statement of findings as well as deepen our understanding of what will be required before others will feel ready to incorporate what we have learned into their own teaching. The innovation and implementation of the backtable bots was received with a lot of excitement and several of our colleagues expressed an interest in using our bots in their teaching. We were clear that developing their own prompts with their own specific pedagogy objectives and preferences would require properly skilled staff support and a commitment to a substantial volume of work.

Figure 16

Excerpts from Student–Backtable Bot Conversation

 **Helper Bot**

Hello Dana, it's Martin Montoya here, Chief of Staff for the Mayor of Villaverde. I remember our university days fondly. How can I assist you today regarding the hydropower station discussions?

4/30/2024 09:22 AM

Hi Martin, nice to see you again! The hydropower station is going to bring some big changes to villaverde as a whole, including with the river's water supply and the construction around the city and the influx of workers. Do you have any concerns about the initiative? What are some benefits you foresee?

4/30/2024 09:25 AM

 **Helper Bot**

Hello Dana, it's great to reconnect. The Mayor prioritizes a balance between development and environmental preservation. Benefits we anticipate include infrastructure improvements and potential job creation. What key concerns do the residents of Villaverde have about the project?

4/30/2024 09:25 AM

The residents are worried about the tourism that may be affected if the river dries up and the loud disturbances from the construction. How does the mayor's office feel about the tourism in villaverde?

4/30/2024 09:26 AM

 **Helper Bot**

Understandably, tourism is essential for Villaverde's economy. The Mayor's office is attentive to these concerns and seeks to ensure that development projects, like the hydroelectric plant, do not negatively impact this sector. Are there specific project criteria that could make the community more inclined to support the project?

4/30/2024 09:26 AM

If the community had more guarantees about other initiatives that would benefit them and the economy such as road networks connecting to other areas or investment in local schools, they would feel a lot better.

What does the mayor plan to do to ensure the development is balanced with ecology?

4/30/2024 09:28 AM

 **Helper Bot**

The Mayor is keen on ensuring infrastructure benefits, like improved transport and education, align with community needs. Engaging in transparent consultations to incorporate community feedback is a priority. On the ecological side, the Mayor supports environmental safeguards. How does the community view the potential for the project to contribute to economic development, considering these environmental considerations?

4/30/2024 09:28 AM

The community thinks there could be a lot of economic benefit in terms of worker influx and increased tourism for the hydroplant, but are worried the job opportunities will not remain local

4/30/2024 09:29 AM

Figure 16

(Continued)

 **Helper Bot**

It's important that job opportunities benefit local residents. The Mayor is committed to advocating for this. Can you share how the Junta de Vecinos plans to engage with residents who are currently undecided or opposed to the project?

4/30/2024 09:29 AM

We plan on emphasizing the benefits the project will bring for the community and will try to invest more in the areas where residents are more undecided

4/30/2024 09:30 AM

 **Helper Bot**

That sounds like a constructive approach. Engaging residents in dialogue is crucial. What are the most critical environmental impacts the community is worried about, and what mitigation measures would they like to see?

4/30/2024 09:30 AM

They are worried the river will dry up

4/30/2024 09:30 AM

 **Helper Bot**

Preserving the river's flow is indeed a priority. I'll convey these environmental concerns to the Mayor. Is there anything else you'd like to discuss, Dana?

4/30/2024 09:30 AM

What is the biggest concern the mayor has for this project?

4/30/2024 09:30 AM

 **Helper Bot**

The Mayor's primary concern is ensuring that the project delivers tangible benefits without major negative impacts on Villaverde's environment and tourism. Do you feel we're aligned on goals for the community and this project?

4/30/2024 09:30 AM

I think so. One more question off the table, do you think the mayor cares more about the ecological impacts or the infrastructure benefits and development of the plant?

4/30/2024 09:32 AM

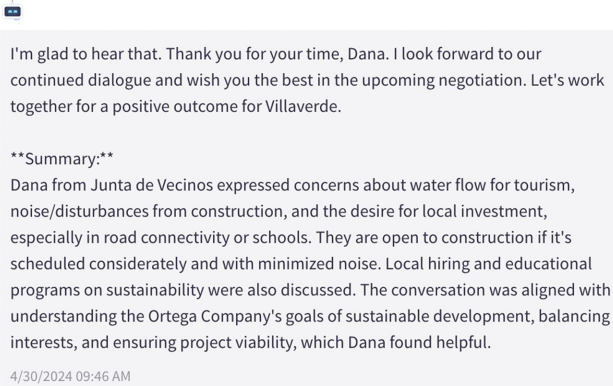
 **Helper Bot**

The Mayor truly values both; it's about achieving a sustainable balance. The vision is for infrastructure benefits to enhance Villaverde without compromising ecological integrity. Thank you for this dialogue, Dana. I wish you well in the upcoming negotiations. Let's stay in touch.

4/30/2024 09:32 AM

Figure 17

Excerpts of Closing and Conversation Summary by a Backtable Bot



I'm glad to hear that. Thank you for your time, Dana. I look forward to our continued dialogue and wish you the best in the upcoming negotiation. Let's work together for a positive outcome for Villaverde.

****Summary:****

Dana from Junta de Vecinos expressed concerns about water flow for tourism, noise/disturbances from construction, and the desire for local investment, especially in road connectivity or schools. They are open to construction if it's scheduled considerably and with minimized noise. Local hiring and educational programs on sustainability were also discussed. The conversation was aligned with understanding the Ortega Company's goals of sustainable development, balancing interests, and ensuring project viability, which Dana found helpful.

4/30/2024 09:46 AM

Recently we had one colleague successfully use our Harborco backtable bots in her undergraduate negotiation classes (28 and 23 students). Engagement with the bots was optional and the students who did engage generally said they found it useful. When asked why, they said it helped them (a) better gauge the other parties' priorities; and (b) better assess who might be a promising coalition partner. A couple of students said that they didn't find the bot useful because "it wouldn't tell them anything specific." However, in retrospect, it appears that those students didn't share any information with the bot, and therefore the bot—who was instructed to behave reciprocally and not share any information if none is shared—behaved realistically (as designed, and also as a human being would probably behave in the real world). Several students said that they found that the backtable bot conversation helped them learn how to reword their questions. In addition to monitoring the students' interactions, the instructor herself experienced the simulation and commented that she found the bots surprisingly realistic in their responses. Clearly, this is still a small sample, but another encouraging sign from a very experienced negotiation instructor.²³

Implications of Our Findings

Our pilot experience suggests that it is possible to develop and implement tools using GenAI for multistakeholder negotiation coaching. We also feel that rapid developments in GenAI technology will make it possible in the very near future to develop even better tools to

accomplish our instructional goals.²⁴ Current bot interactions with learners (including the preparation bot, the backtable bots, and the debriefing bot) used back-and-forth text messaging. New technology is recently available that can convert speech to text and text to speech. This will enable students to interact with their coaching bots in normal natural language conversation. This could include conversations with an avatar that displays emotive facial expressions and tone, and can be complemented by interpretation of student facial expressions and tone by the coaching bot.

The addition of backtable bot interactions has demonstrated how a one-session simulation can be expanded into a multiphase experience, while retaining the logistics of a one-session in-class in-person negotiation session. The fact that this step is asynchronous (done at a time of the student's own choosing without any prior coordination) increases the convenience for the teaching faculty²⁵ as well as the time and intensity of student engagement.

There are some in academia who want to see for themselves what are the impacts of negotiation bots. The issue of how to measure the effectiveness of teaching key negotiation concepts and methods using particular role-play simulations has been debated for quite some time (including recently).²⁶ The discussion will escalate as we try to assess the added effects of incorporating Gen AI assistants.

We are eager to continue to explore and broaden the use of negotiation bots in several ways, and encourage others to do the same, based on the following beliefs:

1. The backtable bot concept can be applied to two-party negotiation as well as multiparty negotiation, even in introductory courses, to encourage students to include in their preparation a "reaching out to someone at the other party" to learn more about the other side.
2. Incorporating backtable bots in various ways into existing role-play simulations (designed for one-session in-person negotiation) provides richer (more three-dimensional) asynchronous experiences.²⁷
3. Using negotiation coaching bots can emphasize additional elements of negotiation instruction in multiparty situations, particularly the roles of facilitators and mediators.
4. Using negotiation coaching bots that carry documentation ("the memory") of the interaction with a student from one exercise to the next can enable extended reflections on the evolution of the learning over a full semester.
5. GenAI bots can similarly be developed to help students in courses other than those on negotiation, such as leadership and entrepreneurship classes.

Conclusion

Our preliminary experiments have shown that negotiation bots can be used to enhance professional instruction in a number of ways. Our work suggests that negotiation coaching bots can be used to enhance student learning, both before and after engaging in simulated role-play negotiations; and negotiation backtable bots can be added to enhance existing role-play simulations, expanding the dimensions of negotiation interaction that are possible. With current and upcoming improved capabilities and greater accessibility of GenAI, we believe it will be possible for college instructors and professional trainers to use GenAI-assisted tools to enhance how people learn to negotiate, and how they can learn from their own experience.

Notes

1. An earlier draft of this article was published as a preprint in a collection of papers reporting on AI research at the Massachusetts Institute of Technology (Susskind et al. 2024).
2. We used the online learning platform called iDecisionGames, referred to herein as IDG, which refers to the learning platform or the learning system. See <https://idecisiongames.com/promo-customers>. We want to thank Niraj Kumar and the team at IDG for their collaboration and support.
3. For more on 3-D negotiation and treating a negotiation as a series of negotiations, see Lax and Sebenius (2006).
4. Our work this past semester (spring 2024) involved building 6 individual preparation coaching bots for each of the 6 roles (in each simulation) and 6 pre-negotiation backtable coaching bots, each with specific instructions on how to interact with the 5 other roles (i.e., 30 possible pairs of back table interactions for each simulation).
5. The learning system records all student interactions, so instructors are able to review each student's levels of engagement with, and understanding of, all the course materials, as well as the quality of the coaching bot interactions.
6. Unlike static tools like calculators, which call for a purely functional relationship between user and technology, our coaches, by displaying unique characteristics and behaviors, are able to foster more dynamic and invested interactions.
7. Prompt design is the key to building effective negotiation coaching bots. We developed a four-aspect lens to define what we needed from each bot prompt. We named these aspects: substance, style, session, and memory. While we were using ChatGPT-4 Turbo, these concepts translate to any other LLM that might be used as the back end of the system. Naturally, if switching LLMs (or upgrading to a new version of an existing one), we would want to test the system to make sure that the "behaviors" have not been modified materially.
8. Ensuring all students arrive on time and prepared is a common challenge when teaching multiparty negotiation, since missing one student ruins the learning opportunity for the entire group. If a student is absent or running late, we, as the instructors, typically need to insert an alternate student (or a teaching assistant) who has prepared fully for the role to ensure that the group is able to have a worthwhile learning experience.
9. We are grateful for graduate student course assistants Swati Garg and Yujie Wang for their support.

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10. gpt-4-0125-preview from <https://platform.openai.com/docs/models/o1>
 11. The experiments were first run with two groups in Susskind's class and lessons from the first run were implemented before the games were run with three groups in Dinnar's class.
 12. Because this was the first time the simulations with bots were used, we established a baseline measuring the participants' assessments of their own negotiating capabilities before using AI.
 13. Some instructors may still elect to include such an assignment but can now choose to do it before or after steps C or D.
 14. All the students needed to have access to their laptops and the IDG platform during their in-person negotiations in case they needed to review their previous bot conversations, as well as to record their votes/results.
 15. Some groups had two people play one role together, negotiating as a team representing the one role. In such a case, the other students would receive a back table interactions summary indicating that both students had separate interactions with their backtable bot, and two summaries would be provided.
 16. Each role-play simulation typically was conducted across two lecture/lab sessions. The preparations happened before the first session; in-person negotiations happened during the first session; the stages of debriefs and reflections were deployed after the negotiation session; and finally, all the material from prior stages was used in the in-class debrief during the second class.
 17. We ran all exercises with both coaching and backtable bots. We did not run a control group with only coaching bots.
 18. Moreover, 100% of our students surveyed answered that they found the preparation coaching bots extremely beneficial—pushing them to prepare more effectively.
 19. The preparation bot was particularly valuable in getting students to consider the interests of all parties, suggesting creative solutions, providing a systematic framework for preparation, and allowing private conversations with a bot to explore information regarding the pressures being put on their negotiating partners by their back tables.
 20. Given the asynchronous nature of bot interactions, we implemented a 30-minute time-out threshold to gauge student engagement more accurately. This time-out allowed us to differentiate between active engagement and periods when students might have stepped away from their devices. If a gap exceeding 30 minutes occurred between a bot's message and a student's response, that interval was excluded from the total interaction time for that session.
 21. A few students still reported being too busy to take full advantage, but nearly all took advantage of this asynchronous opportunity to conduct these conversations at their own convenient time and without any prior planning (as would be required if they were offered the opportunity to negotiate with another student).
 22. The Pedagogy at PON initiative is dedicated to improving the way people teach and learn about negotiation and dispute resolution. It serves as an intellectual focal point for peer discussions of negotiation research, curriculum development, training, and networking. https://www.pon.harvard.edu/category/research_projects/negotiation-pedagogy-program-on-negotiation/
 23. We thank Melissa Manwaring, Associate Professor of Practice at Babson College, for her collaboration and feedback.
 24. As we finalized the semester and the writing of this article, OpenAI announced the availability of more advanced GPT versions, which promise some exciting innovations. Other LLM providers did as well. These tools will continue to evolve quickly.
 25. Integration into the flexible learning platform also allows instructors to make group changes up to the last minute, reducing staff workload and learning delays due to unexpected group reassignments (due to unexpectedly absent students).

26. Negotiation Pedagogy Forum at PON: Evaluating Teaching and Mentoring Performance, February 6, 2023, Harvard University.
27. Naturally, future new role-play simulations could be developed from the start with negotiation bots (both coaching bots and backtable bots).

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